

Venus Labs - PR Fix

Security Assessment

CertiK Assessed on Mar 20th, 2026





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Venus Labs - PR Fix

The security assessment was prepared by Certik.

Executive Summary

TYPES

Lending

ECOSYSTEM

Binance Smart Chain
(BSC)

METHODS

Manual Review, Static Analysis

LANGUAGE

Solidity

TIMELINE

Preliminary comments published on 03/18/2026

Final report published on 03/20/2026

Vulnerability Summary



2

Total Findings

1

Resolved

0

Partially Resolved

1

Acknowledged

0

Declined

1 Centralization

1 Acknowledged



Centralization findings highlight privileged roles & functions and their capabilities, or instances where the project takes custody of users' assets.

0 Critical

Critical risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.

0 Major

Major risks may include logical errors that, under specific circumstances, could result in fund losses or loss of project control.

1 Medium

1 Resolved



Medium risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform.

0 Minor

Minor risks can be any of the above, but on a smaller scale. They generally do not compromise the overall integrity of the project, but they may be less efficient than other solutions.

0 Informational

Informational errors are often recommendations to improve the style of the code or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

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Disclaimer

CODEBASE | VENUS LABS - PR FIX

Repository

<https://github.com/VenusProtocol/venus-protocol/pull/664>

<https://github.com/VenusProtocol/isolated-pools/pull/551>

Commit

[19c1ac92e1021584a7158f6e6236c0ff353604a4](#)

[0a905ab9ff2b3c8714dfa851dc9ff4b1e80d2729](#)

[2882bc532fbdd619114357b413caf2edc9ae3ac4](#)

AUDIT SCOPE | VENUS LABS - PR FIX

VenusProtocol/venus-protocol

 VBep20.sol

 VTokenInterfaces.sol

VenusProtocol/isolated-pools

 VToken.sol

 VTokenInterfaces.sol

APPROACH & METHODS | VENUS LABS - PR FIX

This audit was conducted for Venus Labs to evaluate the security and correctness of the smart contracts associated with the Venus Labs - PR Fix project. The assessment included a comprehensive review of the in-scope smart contracts. The audit was performed using a combination of Manual Review and Static Analysis.

The review process emphasized the following areas:

- Architecture review and threat modeling to understand systemic risks and identify design-level flaws.
- Identification of vulnerabilities through both common and edge-case attack vectors.
- Manual verification of contract logic to ensure alignment with intended design and business requirements.
- Dynamic testing to validate runtime behavior and assess execution risks.
- Assessment of code quality and maintainability, including adherence to current best practices and industry standards.

The audit resulted in findings categorized across multiple severity levels, from informational to critical. To enhance the project's security and long-term robustness, we recommend addressing the identified issues and considering the following general improvements:

- Improve code readability and maintainability by adopting a clean architectural pattern and modular design.
- Strengthen testing coverage, including unit and integration tests for key functionalities and edge cases.
- Maintain meaningful inline comments and documentations.
- Implement clear and transparent documentation for privileged roles and sensitive protocol operations.
- Regularly review and simulate contract behavior against newly emerging attack vectors.

FINDINGS | VENUS LABS - PR FIX



2

Total Findings

0

Critical

1

Centralization

0

Major

1

Medium

0

Minor

0

Informational

This report has been prepared for Venus Labs to identify potential vulnerabilities and security issues within the reviewed codebase. During the course of the audit, a total of 2 issues were identified. Leveraging a combination of Manual Review & Static Analysis the following findings were uncovered:

ID	Title	Category	Severity	Status
VLP-01	Admin Can Sweep Arbitrary Tokens Via <code>sweepTokenAndSync()</code>	Centralization	Centralization	● Acknowledged
VLP-02	<code>VToken.badDebtRecovered()</code> Doesn't Update <code>internalCash</code>	Logical Issue	Medium	● Resolved

VLP-01 | Admin Can Sweep Arbitrary Tokens Via `sweepTokenAndSync()`

Category	Severity	Location	Status
Centralization	● Centralization	VBep20.sol (PR664): 263~265	● Acknowledged

Description

The `sweepTokenAndSync()` function in `VBep20.sol` allows the admin to transfer any amount of the underlying token to themselves, effectively enabling token theft from the contract.

```
function sweepTokenAndSync(uint256 transferAmount) external {
    require(msg.sender == admin);

    if (transferAmount > 0) {
        IERC20(underlying).safeTransfer(msg.sender, transferAmount);
        emit TokenSwept(msg.sender, transferAmount);
    }

    uint256 oldInternalCash = internalCash;
    internalCash = IERC20(underlying).balanceOf(address(this));
    emit CashSynced(oldInternalCash, internalCash);
}
```

The function lacks any validation that `transferAmount` is actually "excess" or that it doesn't exceed the contract's legitimate balance. While it updates `internalCash` after the transfer, this doesn't prevent the admin from draining funds that belong to users (depositors). This creates a centralization risk where a malicious or compromised admin could steal user funds.

Recommendation

Restrict `sweepTokenAndSync()` function to only transfer tokens that are genuinely excess (e.g., tokens that were accidentally sent to the contract and are not part of `internalCash`).

Alleviation

[Venus Labs, 03/20/2026]: `sweepTokenAndSync()` is an admin functionality, and we make sure the transfer tokens are used correctly.

VLP-02 | VToken.badDebtRecovered() Doesn't Update internalCash

Category	Severity	Location	Status
Logical Issue	● Medium	VToken.sol (PR551): 659~661	● Resolved

Description

The `VToken.badDebtRecovered()` function reduces the `badDebt` storage variable when bad debt is recovered from an auction, but it does not update the `internalCash` balance. This creates an inconsistency between the tracked internal cash and the actual token balance of the contract.

When `Shortfall.closeAuction()` transfers underlying tokens to the VToken contract to repay bad debt, the actual token balance increases, but `internalCash` remains unchanged. This discrepancy can affect calculations that rely on `internalCash`, such as `_getCashPrior()` and `_exchangeRateStored()`.

In `Shortfall.sol`, the `closeAuction()` function transfers underlying tokens to VToken markets:

```
uint256 balanceBefore = erc20.balanceOf(address(auction.markets[i]));
erc20.safeTransfer(address(auction.markets[i]),
    auction.bidAmount[auction.markets[i]]);
uint256 balanceAfter = erc20.balanceOf(address(auction.markets[i]));
marketsDebt[i] = balanceAfter - balanceBefore;

auction.markets[i].badDebtRecovered(marketsDebt[i]);
```

Recommendation

Update `internalCash` in `badDebtRecovered()` to reflect the incoming tokens. Add `internalCash += recoveredAmount_;` after the validation checks.

Alleviation

[Venus Labs, 03/20/2026]: The team heeded the advice and resolved the issue in

<https://github.com/VenusProtocol/isolated-pools/pull/551/changes/a48ea616ce80485ea0373e3dcbfb47f82398f776>

APPENDIX | VENUS LABS - PR FIX

Finding Categories

Categories	Description
Centralization	Centralization findings detail the design choices of designating privileged roles or other centralized controls over the code.
Logical Issue	Logical Issue findings indicate general implementation issues related to the program logic.

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